Team Deuterium · all written reports

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Results report (results.md)

QFF2026 QML Track — Results Report

Team name: Deuterium **Members:** Charan Reddy, Krishna Neeraj, Ismail Mohammad **Selected PS:** [x] PS-Q1 [ ] PS-Q2 [ ] PS-Q3
Date: 2026-10-30 **Dependency versions:** pennylane==0.44.1, PennyLane-qiskit==0.44.1, qiskit==2.3.0, qiskit-aer==0.17.2, scikit-learn==1.9.1, numpy==2.4.6, pandas==3.1.0 (Python 3.13.13; full list in README.md) **Noise model version:** QFF2026-v2
All numbers below are produced by `solution.ipynb` (and, identically, by `vqc_work/scripts/train_vqc.py`, which was run twice from scratch with identical results). All PS-Q1 metrics are on the validation set, which was also used for model selection, so they are optimistic; the hidden test set was used only for the final predictions.

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1. Approach Summary

We classify heartbeats with a 4-qubit hybrid classifier: the 30 window statistics are standardised and reduced to 4 principal components (fitted on training rows only), each component is rescaled to [-π, π] with a training-fitted scale and loaded as an `RY` angle, two trainable layers of `RY`+`RZ` rotations with a bidirectional CX ring (CX(0,1) CX(1,0) CX(1,2) CX(2,1) CX(2,3) CX(3,2) CX(3,0) CX(0,3)) process it, and $\langle Z_0 \rangle$ goes through a trainable affine map and a sigmoid. The angle rescaling is the key design decision: raw PCA scores reach ±8.6 rad and 52.8% of training rows have an angle outside [-π, π], so the 2π-periodic `RY` wraps distant samples together; rescaling lifts validation macro-F1 from 0.9633 to 0.9993 for the same circuit. The bidirectional 4-qubit ring is matched by our Processor D, so the circuit runs there without routing. We also analysed what the ansatz can learn: the final ring maps Z_0 to $Z_0Z_2Z_3$, so 6 of the 16 rotation angles never receive a gradient and a 1-layer version cannot see PC2 at all, which is why two layers are needed.

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2. Results

Classical baseline

Model	Official metric	Notes
Logistic regression, 30 standardised features	macro-F1 **1.0000** (validation, 0 errors)	Grid C ∈ {0.01, 0.1, 1, 10, 100} × class_weight ∈ {None, balanced}, all 10 points give 1.0000; selected C=1, no weighting; scaler fit on train only, selection on validation only

Quantum / hybrid model

Official metric: Q1 macro-F1. Ideal = exact statevector; noisy = official QFF2026-v2 model, `prepare_for_noisy_run()` on every executed circuit, 8192 shots, simulator seeds 42–46 (mean ± sd).

Configuration	Ideal	Noisy (benchmark)	Notes
Submitted: scaled angles, 2 layers, lr 0.001, epoch 19	**0.9993** (95% bootstrap CI 0.9979–1.0000; 1 error of 1500)	0.4663 ± 0.0048	Same 0.9993 for all 5 training seeds. Noise shrinks $\langle Z_0 \rangle$ to 0.40× but keeps the ranking (ROC-AUC 1.0000 ideal and noisy); the fixed threshold then calls most Normal beats Arrhythmia (noisy recall: Normal 0.192, Arrhythmia 1.000)
Ablation: direct (unscaled) angles, 2 layers, lr 0.01	0.9633 (CI 0.9534–0.9727)	0.7971 ± 0.0037	5 training seeds: 0.9636 ± 0.0008
Diagnostic: scaled angles, 1 layer, lr 0.01	0.9667 (CI 0.9570–0.9751)	0.9496 ± 0.0013	8 CX instead of 16: shrinkage 0.60× instead of 0.40×, far more robust; cannot use PC2
Predeclared grid, all 8 runs	scaled: 0.9667 / 0.9660 / 0.9993 / 0.9993; direct: 0.8855 / 0.8806 / 0.9633 / 0.9511	—	(1 layer lr 0.01 / 1 layer lr 0.001 / 2 layers lr 0.01 / 2 layers lr 0.001)

Test predictions (`predictions_q1.csv`) come from the submitted model under ideal simulation. They agree with the classical baseline's test predictions on 1500 of 1500 samples (a sanity check; no test labels are available to us).

Noise mitigation (PS-Q3 required; optional for PS-Q1/Q2)

Strategy	Mitigated metric	Extra circuit evals	Justified?
Readout recalibration: re-fit only the 2 readout parameters (w, b) on noisy validation $\langle Z_0 \rangle$, circuit weights fixed; scored out-of-fold (5-fold)	Submitted model 0.4731 → **0.9986**; direct ablation 0.7993 → 0.9612; 1-layer 0.9487 → 0.9640	None beyond one noisy pass over the 1500 validation circuits (8192 shots each)	Yes, because noise mostly rescales $\langle Z_0 \rangle$ (correlation 0.998 with the ideal value). It needs labelled calibration data and assumes the device noise stays the same. Reported as an analysis only; it is not used for the predictions

3. Task A — Bell state |Φ+⟩ (required)

Produced by `task_a_bell()` in `benchmark/qff_qml_benchmark.py` (8192 shots per basis).

Fidelity definition used: F = (1 + ⟨XX⟩ − ⟨YY⟩ + ⟨ZZ⟩) / 4, with ⟨P⟩ = (N_even − N_odd)/N, SE(⟨P⟩) = √((1 − ⟨P⟩²)/N) and SE(F) = √(SE_XX² + SE_YY² + SE_ZZ²)/4. (P(00)+P(11) is a population, not a fidelity, and is not accepted.)

Condition	⟨XX⟩ ± SE	⟨YY⟩ ± SE	⟨ZZ⟩ ± SE	F ± SE
Ideal	+1.0000 ± 0.0000	−1.0000 ± 0.0000	+1.0000 ± 0.0000	1.0000 ± 0.0000
Noisy (benchmark)	+0.8191 ± 0.0063	−0.8279 ± 0.0062	+0.8735 ± 0.0054	0.8801 ± 0.0026

What the difference says about noise and entanglement: In the ideal run all three correlators are exactly ±1, the signature of a pure |Φ+⟩. Depolarizing errors on the H and CX gates and 2% readout flips mix the state toward the maximally mixed state, so every correlator shrinks and F drops to 0.880. ZZ (0.874) degrades least because its circuit has no basis-change gates, while XX and YY pass through extra single-qubit gates (and S† for YY) before measurement. F remains far above 0.5, the value of a classical 50/50 mixture of |00⟩ and |11⟩, so the entanglement survives; but about 12% of the overlap is lost after a single CX, which foreshadows the much larger loss of our 16-CX classifier.

4. Processor D Benchmark (Task B)

Produced by `python benchmark/qff_qml_benchmark.py processor_d_config.json`.

QFT-3 fidelity definition used: mean over the 8 computational-basis inputs |j⟩ of F_j = ⟨ψ_j|ρ_j|ψ_j⟩, where ψ_j = QFT|j⟩ (ideal) and ρ_j is the density matrix of the 3 logical qubits after the transpiled circuit runs under the benchmark noise model (gate noise; no measurement). Ideal must be 1.0.

Processor ID: qff_team_ring4_chord_v1 **n_qubits:** 4 **Topology:** ring 0-1-2-3-0 plus chord 0-2, all links two-way; native gates rx, ry, rz, h, cx

Benchmark	Value
QFT-3 depth (before transpile)	6
QFT-3 depth (after transpile)	16
Two-qubit gate count	9
SWAP overhead	0
Fidelity (ideal)	1.0
Fidelity (noisy)	0.677499
Coupling map direction (one-way / two-way links)	two-way (all 5 links listed in both directions, 10 directed pairs)

Design comparison (all measured with the official `task_b_qft3`):

Design	Depth after	2q gates	SWAPs	Noisy QFT-3 F
Linear 5, one-way links	31	12	1	0.5504
Linear 5, two-way links (organizer example)	22	12	1	0.6036
Ring 4, two-way	19	12	1	0.6351
Ring 4 + chord 0-2, two-way, no native h	18	9	0	0.6684
Ring 4 + chord 0-2, two-way, native h (submitted)	**16**	**9**	**0**	**0.6775**

What do the results tell you about your topology? Direction matters: listing the same line one-way instead of two-way costs 9 extra layers of depth and 0.053 fidelity, because every reversed CX needs extra single-qubit gates. QFT-3 needs all three qubit pairs to interact, so a plain ring pays one SWAP (3 extra CX); the single chord 0-2 closes a triangle, removing the SWAP and reaching the 9-two-qubit-gate minimum, and native `h` trims depth from 18 to 16. For our own circuit the 4-qubit ring is exact: the 2-layer classifier keeps its 16 CX at depth 22 on Processor D versus 34 CX at depth 36 on the linear example, and since our noise analysis shows the classifier's accuracy falls with every CX, avoiding routing directly matters. The chord is one extra coupler (crosstalk and fabrication cost on real hardware) that the benchmark does not penalise; a second chord gave no further gain, so we stopped at one.

5. Resource Costs

Resource	Value
Qubits used	4
Circuit depth	Logical 21 (16 CX); after `prepare_for_noisy_run()` 21 with 16 single-qubit gates and 16 CX; on Processor D 22 with 16 CX

Resource	Value
Trainable parameters	18 nominal (16 rotation angles + readout w, b); 12 effective (6 angles have structurally zero gradient)
Training time (approx)	30 epochs: about 140 s (1 layer) and 210 s (2 layers) per run while 8 runs share a 10-core CPU; the full notebook runs in about 10 minutes
Total circuit evaluations	Training: 405,000 forward evaluations per run (375 batches × 16 samples + full train/validation scoring per epoch × 30 epochs), 6,480,000 for all 16 runs; gradients by backpropagation through the simulator, so no extra circuit executions. Noisy evaluation: 1,500 circuits × 8,192 shots × 5 seeds = 61.44 M shots per model

6. Limitations and Honest Assessment

- **Quantum is not competitive with classical here.** Logistic regression on the 30 features is perfect on validation (and the organizers' baseline is 1.000 on the hidden test set), with no quantum hardware, no noise and fewer assumptions. Our classifier matches it only in ideal simulation (one validation error) and only after the angle-scaling fix. We claim no quantum advantage. This dataset is near-linearly separable, which a 4-qubit circuit can represent but has no reason to beat.
- **Noise is the real limitation.** Under QFF2026-v2 the submitted model falls to 0.466 macro-F1. The cause is understood: noise attenuates $\langle Z_0 \rangle$ (0.40×) while preserving the ordering, and the decision threshold learned in ideal training no longer fits. A shallower circuit (1 layer, 8 CX) is far more robust (0.950) but weaker ideally and blind to PC2; recalibrating the readout on noisy validation data recovers 0.999. Any real deployment would need that kind of noise-aware calibration.
- **Selection bias.** Validation chose the epoch, configuration and encoding, and the scaled encoding was introduced after inspecting the direct encoding's behaviour; validation scores are therefore optimistic.
- **Ansatz redundancy.** The specified ansatz has dead parameters (all four RZ just before the final ring, RY on qubit 1 in the last layer, and RZ on qubit 2 in the first layer); a leaner ansatz could reach the same result with fewer gates.
- **Synthetic data.** Features are modelled on MIT-BIH statistics; no clinical validity is claimed.
- **Process note.** Early in the session, a shell row count (`wc -l`) over the q1 CSVs touched the test-features file; no content was read or used for modelling. The test features are read only in Section 10 of the notebook, for inference.

7. Reproducibility

```
SEED = 42              # numpy, random, PCA, weight init and batch order, Aer simulator, transpiler
EXTRA_SEEDS = [43, 44, 45, 46]        # training-robustness reruns of the selected configurations
NOISY_SIM_SEEDS = [42, 43, 44, 45, 46]  # simulator seeds for the noisy evaluation
```

- Two full independent runs of `vqc_work/scripts/train_vqc.py` produced identical configs, 480-row training histories and resource tables (all non-timing fields).
- Retraining the submitted configuration reproduces its parameters exactly (max difference 0.0); reloading the saved JSON model reproduces validation probabilities to 5.6e-16.
- `python vqc_work/scripts/train_vqc.py --selftest` runs 12 checks: macro-F1 matches scikit-learn, BCE loss, gradients match finite differences, partial last batch, the CX-ring identity, exact Qiskit/PennyLane equivalence before and after transpilation, noised-basis coverage, bit ordering, readout noise, signal attenuation, and a guard that only the training/validation files are read.
- The noisy pipeline is verified in the notebook: X on qubit 0 gives $\langle Z_0 \rangle = -1$ and X on qubit 1 gives +1; an empty circuit reads 1 with probability 0.0181 (expected 0.02); executed circuits equal the trained circuit to 1e-15 before noise.

8. Optional Extensions

[ ] Second mitigation strategy (PS-Q3) [x] Ablation studies: angle encoding (scaled vs direct), depth (1 vs 2 layers), 5 training seeds, 5 simulator seeds, structural (zero-gradient) analysis of the ansatz [x] Other: readout recalibration under noise (Section 2), Processor D design comparison (Section 4)

README

QFF2026 QML Track - Phase 2: PS-Q1

Problem statement: PS-Q1 (ECG heartbeat classification, QML-BIO-MITBIH). Common tasks: **Task A** (Bell state) and **Task B** (Processor D). **Team:** Deuterium **Members:** Charan Reddy, Krishna Neeraj, Ismail Mohammad

(Same team and problem statement as Phase 1.)

Everything runs from `solution.ipynb`: Task A, Task B, the classical baseline, the hybrid quantum classifier (training, ideal and noisy evaluation) and the test predictions. The full report is `results.md`.

Setup (Python 3.11 - 3.13)

```
python3 -m venv qml-env
source qml-env/bin/activate            # Windows: qml-env\Scripts\activate
python -m pip install --upgrade pip
pip install -r requirements.txt        # the organizer's pinned kit v3 versions
```

Run

All commands are run from the repository root.

```
python noise_model/benchmark_noise_model.py
```

```
jupyter nbconvert --to notebook --execute --inplace --ExecutePreprocessor.timeout=-1 solution.ipynb
```

The notebook takes about 10 minutes on a 10-core laptop (training runs in parallel processes; allow 15-30 minutes on fewer cores). It rewrites `predictions_q1.csv`. Opening it with `jupyter notebook solution.ipynb` and running all cells works too.

Check the deliverables:

```
python tools/validate_predictions.py q1 predictions_q1.csv
```

```
python benchmark/qff_qml_benchmark.py processor_d_config.json
```

Optional: regenerate the VQC reports and artifacts in `vqc_work/` (about 11 minutes), or run its fast self-test (seconds):

```
python vqc_work/scripts/train_vqc.py --selftest
```

```
python vqc_work/scripts/train_vqc.py
```

Files

Path	What it is
`solution.ipynb`	Runnable notebook: PS-Q1 + Task A + Task B, with outputs
`processor_d_config.json`	Processor D design; `benchmark_results` equal the official re-run
`predictions_q1.csv`	Test predictions (`sample_id,label`), passes `tools/validate_predictions.py`
`results.md`	Report following the organizer's `results_template.md` (kit v3)
`vqc_work/scripts/train_vqc.py`	Quantum-model code used by the notebook; standalone pipeline and `--selftest`
`vqc_work/reports/`, `vqc_work/artifacts/`	Detailed VQC reports, resource summary, saved model state and training history
`scripts/train_classical_baseline.py`, `reports/classical_baseline_metrics.csv`	Classical logistic-regression baseline (validation-tuned) and its grid results
`data/q1/`, `noise_model/`, `benchmark/`, `tools/`, `schemas/`, `requirements.txt`	Unmodified organizer kit v3 files needed to run (checked against the kit's MANIFEST.sha256)

Rules followed

- Only PS-Q1 data is used. Scaler, PCA and the angle scale are fitted on `ecg_train.csv` only; `ecg_val.csv` is used for model selection only; the test features are read once, for inference, after the model is fixed.

- Every noisy run uses `noise_model/benchmark_noise_model.py` unchanged, with `prepare_for_noisy_run()` applied to each executed circuit (basis `rx, ry, rz, cx`).
- Task A and Task B numbers come from `benchmark/qff_qml_benchmark.py`.
- Seed 42 everywhere (plus documented extra seeds 43-46 for robustness checks).

Environment (`pip freeze`)

Produced with Python 3.13.13 in a venv built from `requirements.txt` (also saved as `pip_freeze.txt`):

```
annotated-types==0.8.0
anyio==4.15.1
appdirs==1.4.4
appnope==1.0.0
argon2-cffi==25.1.0
argon2-cffi-bindings==26.1.0
arrow==1.4.0
asttokens==3.0.2
astunparse==1.6.3
async-lru==2.4.0
attrs==26.1.0
autograd==1.9.1
autoray==0.8.2
babel==2.18.0
beautifulsoup4==4.15.0
bleach==6.4.0
cachetools==7.2.7
certifi==2026.10.23
cffi==2.1.1
charset-normalizer==3.5.2
cloudpickle==3.1.2
comm==0.2.3
contourpy==1.4.0
cryptography==50.0.2
cycler==0.12.1
debugpy==1.8.22
defusedxml==0.7.1
diastatic-malt==2.15.2
dill==0.4.1
executing==2.3.0
fastjsonschema==2.22.2
fonttools==4.68.0
fqdn==1.6.0
gast==0.7.0
h11==0.16.0
httpcore==1.0.9
httpx==0.28.1
ibm-cloud-sdk-core==3.28.1
ibm-platform-services==0.77.1
idna==3.20
ipykernel==7.4.0
ipython==9.17.1
ipython_pygments_lexers==1.1.1
isoduration==20.11.0
jedi==0.20.1
Jinja2==3.1.6
joblib==1.6.0
json5==0.17.3
jsonpointer==3.6
jsonschema==4.26.0
jsonschema-specifications==2025.9.1
jupyter-events==0.12.1
jupyter-lsp==2.3.1
```

jupyter_builder==1.2.4
jupyter_client==8.10.0
jupyter_core==5.9.1
jupyter_server==2.21.1
jupyter_server_terminals==0.5.4
jupyterlab==4.6.5
jupyterlab_pygments==0.3.0
jupyterlab_server==2.28.1
kiwisolver==1.5.1
lark==1.3.1
MarkupSafe==3.0.4
matplotlib==3.11.2
matplotlib-inline==0.2.2
mistune==3.4.0
mpmath==1.3.0
narwhals==2.28.0
nbclient==0.11.0
nbconvert==7.17.2
nbformat==5.11.1
nest-asyncio2==1.7.4
networkx==3.7
notebook==7.6.3
notebook_shim==0.2.4
numpy==2.4.6
packaging==26.3
pandas==3.1.0
pandocfilters==1.5.1
parso==0.8.7
pennylane==0.44.1
PennyLane-qiskit==0.44.1
pennylane_lightning==0.45.0
pexpect==4.9.0
pillow==13.0.0
platformdirs==4.14.0
prometheus_client==0.26.0
prompt_toolkit==3.0.53
psutil==7.2.2
ptyprocess==0.7.0
pure_eval==0.2.4
pycparser==3.12
pydantic==2.14.2
pydantic_core==2.50.2
Pygments==2.21.0
PyJWT==2.15.1
pyparsing==3.3.3
pyspnego==0.12.4
python-dateutil==3.0.0
python-json-logger==4.2.0
PyYAML==6.0.3
pyzmq==27.2.0
qiskit==2.3.0
qiskit-aer==0.17.2
qiskit-ibm-runtime==0.45.1
referencing==0.37.0
requests==2.34.2
requests_ntlm==1.3.0
rfc3339-validator==0.1.4
rfc3986-validator==0.1.1
rfc3987-syntax==1.1.0
rpds-py==2026.9.1
rustworkx==0.18.1
scikit-learn==1.9.1

scipy==1.17.1
scipy-openblas32==0.3.34.237.2
Send2Trash==2.1.0
six==1.17.0
soupsieve==3.0.4
stack-data==0.6.3
stevedore==5.9.1
sympy==1.14.0
termcolor==3.3.0
terminado==0.18.1
threadpoolctl==3.7.0
tinycss2==1.5.1
tomlkit==0.15.2
tornado==6.6
traitlets==5.16.1
typing-inspection==0.4.4
typing_extensions==4.16.0
tzdata==2026.6
uri-template==1.3.0
urllib3==2.8.0
wcwidth==0.9.4
webcolors==25.10.0
webencodings==0.6.1
websocket-client==1.9.2
wheel==0.48.0

PS-Q1 VQC - training and ideal validation results

Generated by `vqc_work/scripts/train_vqc.py` (total runtime 10.9 min on 10 cores). Every number below comes from that run.

Challenge context (official documents)

Sources: the participant problem statement (v3), the Phase 2 judging rubric, `noise_model/benchmark_noise_model.py`, `benchmark/qff_qml_benchmark.py`.

- PS-Q1 (QML-BIO-MITBIH): classify heartbeats, 0 = Normal, 1 = Arrhythmia, 30 features, ranking metric macro-F1 (accuracy for reference only).
- Rules used here: train only on `ecg_train.csv`; `ecg_val.csv` for model selection and tuning only; scalers/PCA fitted on training data only; seeds fixed; quantum model trained from scratch; every noisy run with the unchanged QFF2026-v2 helper on circuits transpiled to a fully-noised basis.
- Rubric (100 points): gates G1-G5 (repository, one PS + Task A/B, valid predictions, notebook runs, no test-data fitting), C1 quantum design 30, C2 results 25 (10 from hidden-test macro-F1 via 10 x clip((m - 0.5)/(m_baseline - 0.5), 0, 1) with m_baseline = 1.000 for Q1), C3 noise analysis 20, C4 Processor D 15, C5 reproducibility/report 10, bonus 5 (ablations count). The rubric notes that "Q1 is separated perfectly by a simple classical model, so the judged criteria decide it".
- Judging rounds: the rubric describes independent scoring by at least two judges and re-scoring of large differences; no further round structure is documented.
- Scope of this workstream: the VQC only. Task A/B, test predictions and the classical model are handled elsewhere and are not touched here.

Data

- Train [6000, 31], validation [1500, 31] (30 float features + `label`; these files have no `sample_id`). Labels: train {0: 3616, 1: 2384}, validation {0: 899, 1: 601}.
- Missing 0/0, non-finite 0/0, duplicate rows within train/val 0/0, cross-partition duplicate feature rows 0. Integrity: **passed**.

Preprocessing and encoding

StandardScaler (fit: 6,000 training rows) -> PCA(4, svd_solver="full", random_state=42; fit: scaled training rows) -> angle encoding. Validation rows are only transformed.

component	explained variance ratio	cumulative	angle scale (pi / max train abs)	direct angle range (train)	scaled angle range (val)
PC1	0.4794	0.4794	0.3658	[-8.59, 4.37]	[-2.95, 1.62]
PC2	0.2136	0.6930	0.4035	[-5.85, 7.79]	[-2.25, 2.93]
PC3	0.0703	0.7632	0.4795	[-6.55, 4.83]	[-2.86, 2.73]
PC4	0.0496	0.8128	0.7642	[-4.11, 3.74]	[-3.07, 2.89]

- **Why the encoding was changed.** The originally specified *direct* encoding feeds raw PCA scores into `RY`. Those scores span about +-8.6 rad, and 52.8% of training rows have at least one angle outside [-pi, pi]; because `RY` is 2pi-periodic, distant samples wrap onto similar rotations. The *scaled* encoding multiplies each component by pi / max|train score| (fitted on train only), keeping the map one-to-one on the training range. This change was made explicitly, approved by the team, and the direct encoding is still trained and reported as an ablation. Validation rows outside [-pi, pi] after scaling: 0.00%.
- 4 components keep 81.3% of the scaled-feature variance. An audit diagnostic run outside this script found that a logistic regression on the same 4 training-fitted components also classifies the validation set perfectly, so the 4-component bottleneck does not remove the class information; remaining errors come from the quantum model.

Circuit (verified against the implementation)

- 4 qubits, `RY(angle_i)` on qubit i, then per layer: trainable `RY` x4, trainable `RZ` x4, CX sequence [(0, 1), (1, 0), (1, 2), (2, 1), (2, 3), (3, 2), (3, 0), (0, 3)]. Readout `z = <Z_0>`, `logit = w z + b`, `p = sigmoid(logit)`, fixed threshold 0.5.
- Nominal trainable parameters: 8 per layer + 2 readout (1 layer: 10, 2 layers: 18). Logical resources: 1 layer depth 11, CX 8; 2 layers depth 21, CX 16.
- **Structural finding (gradient-verified).** The final CX ring maps `Z_0` back to `Z_0 Z_2 Z_3` (verified numerically, max error 1.2e-16). Qubit 1 therefore never reaches the readout through the last layer, and `RZ` gates directly before the last ring cannot change a Z-basis outcome. Parameters with exactly zero gradient at every tested point:

| layers | quantum params | dead | effective | dead parameters | mean |d/d angle| for PC1..PC4 | |---|---|---|---|---|---| | 1 | 8 | 5 | 3 | layer0_RY_q1, layer0_RZ_q0, layer0_RZ_q1, layer0_RZ_q2, layer0_RZ_q3 | 0.251, 0.000, 0.248, 0.275 | | 2 | 16 | 6 | 10 | layer0_RZ_q2, layer1_RY_q1, layer1_RZ_q0, layer1_RZ_q1, layer1_RZ_q2, layer1_RZ_q3 | 0.081, 0.101, 0.089, 0.088 |

Consequence: the 1-layer model **ignores PC2 entirely** and its output is a product of cosines of PC1, PC3 and PC4; the 2-layer model uses all four inputs. The architecture is kept as specified (dead parameters are harmless for training, they simply never move); removing them would change the specified ansatz.

Training protocol

Adam (PennyLane `AdamOptimizer`), BCE with logits, batch 16 (375 batches per epoch), 30 epochs, seed 42 for initialisation and batch order, parameters initialised N(0, 0.1^2), readout w = 1, b = 0. Gradients: analytic backpropagation through the statevector simulator. The predeclared grid (layers {1, 2} x lr {0.01, 0.001}) is run for both encodings (231 s wall, 8 runs in parallel). Selection per encoding: highest validation macro-F1, ties -> fewer layers, lr 0.001, earlier epoch.

encoding	layers	lr	trainable	best epoch	best val macro-F1	final train loss	final val loss	train time (s, parallel)	forward circuit evals	
scaled	1	0.01	10	4	0.9667	0.0806	0.0961	155	405000	
scaled	1	0.001	10	19	0.9660	0.0938	0.1042	155	405000	
scaled	2	0.01	18	3	0.9993	0.0003	0.0019	226	405000	
scaled	2	0.001	18	19	0.9993	0.0144	0.0154	224	405000	<- selected
direct	1	0.01	10	6	0.8855	0.3056	0.3135	155	405000	
direct	1	0.001	10	30	0.8806	0.3048	0.3092	154	405000	
direct	2	0.01	18	23	0.9633	0.1276	0.1057	225	405000	<- selected
direct	2	0.001	18	17	0.9511	0.1467	0.1301	226	405000	

Canonical model: scaled_L2_lr0.001_s42 (epoch 19). Validation chose the epoch, the configuration and (after the design change) the encoding, so these are *model-selection* numbers and are optimistically biased; they are not held-out test evidence.

Seed robustness (selected configuration of each encoding; 275 s wall)

encoding	config	best val macro-F1 for seeds 42, 43, 44, 45, 46	best epochs	mean	sd
scaled	L2 lr=0.001	0.9993, 0.9993, 0.9993, 0.9993, 0.9993	19, 20, 18, 20, 20	0.9993	0.0000
direct	L2 lr=0.01	0.9633, 0.9647, 0.9639, 0.9633, 0.9626	23, 19, 25, 20, 15	0.9636	0.0008

Ideal validation results (exact statevector)

model	val macro-F1	95% bootstrap CI	accuracy	ROC-AUC	train macro-F1
canonical: scaled, 2 layer(s)	0.9993	[0.9979, 1.0000]	0.9993	1.0000	0.9997
ablation: direct, 2 layer(s)	0.9633	[0.9534, 0.9727]	0.9647	0.9921	0.9515
diagnostic: scaled, 1 layer	0.9667	[0.9570, 0.9751]	0.9680	0.9940	0.9709

Canonical model, per class:

class	precision	recall	F1
0 Normal	1.0000	0.9989	0.9994
1 Arrhythmia	0.9983	1.0000	0.9992

Confusion matrix (class order [0,1]):

true \ pred	0 Normal	1 Arrhythmia
0 Normal	898	1
1 Arrhythmia	0	601

Predicted P(class 1):

mean	std	min	p5	p25	p50	p75	p95	max	frac predicted 1
0.3910	0.4406	0.0111	0.0136	0.0197	0.0430	0.9278	0.9681	0.9998	0.4013

Comparison with the classical baseline (same data, validation-tuned)

Logistic regression on all 30 standardised features (teammate's grid, read-only from `reports/classical_baseline_metrics.csv`): validation macro-F1 1.0000 (10 of 10 grid points at this value); every validation sample is correct, so its bootstrap interval is [1.0000, 1.0000]. The canonical VQC reaches 0.9993 (1 of 1500 validation samples wrong, 95% CI [0.9979, 1.0000]) from 4 PCA inputs and 18 nominal parameters, under ideal simulation only. It does not beat the classical baseline, and no quantum advantage is claimed: the classical model is simpler, exact and noise-free, while the VQC's validation macro-F1 under the official noise model is 0.4663 (mean of 5 simulator seeds).

Resources

- Qubits 4; nominal trainable 18 (effective 12); logical depth 21; logical CX 16.
- Canonical training run: 224 s (measured while 8 runs shared the CPU). Ideal evaluation of 1,500 validation samples: 26.0 ms.
- Counting convention: "forward circuit evaluations" = input samples pushed through the simulated circuit (training batches plus per-epoch train/validation scoring); 405,000 for the canonical run and 3,240,000 for the 8-run search. Gradient circuit executions are not exposed by backpropagation (no parameter-shift executions). Ideal results use 0 shots. Noisy shot counts are in `vqc_ideal_vs_noisy.md`; all rows are in `vqc_resource_summary.csv`.

Reproducibility and quality checks

| check | status | |---|---| | Only train/val CSVs read (explicit paths; --selftest parses this file and asserts read_csv is only called on TRAIN_CSV, VAL_CSV and the read-only classical metrics file, and that no other ecg_*.csv name appears) | PERFORMED - passed | | Integrity: schema, NaN/inf, labels, duplicate rows within/across partitions | PERFORMED - passed | | Scaler, PCA and angle scale fitted on training rows only | PERFORMED by construction (fit_preprocessing) and code review | | Validation never in gradient updates | PERFORMED by construction: optimizer steps use training batches only; validation only scores epochs | | Gradient correctness (autograd vs finite differences) | PERFORMED in --selftest - passed (max diff < 1e-7) | | Final partial batch handled | PERFORMED in --selftest (37 samples -> 3 batches); not exercised in the real run because 6000 = 375 x 16 | | All losses/parameters/probabilities finite, probabilities in [0,1] | PERFORMED - passed | | Reload from saved JSON reproduces ideal validation probabilities | PERFORMED - max |dp| = 5.55e-16 (tolerance 1e-9): passed | | Exact retrain of the canonical run (seed 42) | PERFORMED - max |dparams| = 0.00e+00, max |dval F1| = 0.00e+00, best epoch match = True | | Seed robustness (4 extra seeds per selected config) | PERFORMED - see table | | Full 8-run search repeated from scratch | NOT RUN (only the canonical configuration was retrained) | | Perfect-score investigation | TRIGGERED (ideal validation macro-F1 > 0.99): no duplicates/overlap, fit scopes train-only, labels as documented; the data is near-linearly separable (the teammate's logistic regression scores 1.000 on validation), so a high score is expected |

Run `python vqc_work/scripts/train_vqc.py --selftest` for the fast checks (metrics, BCE, gradients, partial batch, ring identity, Qiskit/PennyLane equivalence before and after transpilation, basis coverage, bit order, readout noise, data-access guard). Framework versions and all seeds are in `artifacts/vqc_config.json`.

Limitations

- Validation is used for selection, so validation scores are optimistic; there is no held-out estimate inside this workstream.
- The predeclared search is small; one canonical seed is reported, with four extra seeds for robustness.
- The specified ansatz contains structurally dead parameters (see above) and its 1-layer version cannot see PC2.
- The scaled encoding was introduced after inspecting the direct encoding's behaviour; that design step is itself informed by validation results and is disclosed as such.
- Exact (shot-free) ideal results; noisy results include shot noise.
- No claim of quantum advantage or clinical validity; the data are synthetic statistics modelled on MIT-BIH.
- Process note: earlier in the project session a shell row-count (`wc -l` over a glob of the q1 CSVs) touched the unlabelled test-features file; no content was read or used, and this script has no code path to that file (asserted by `--selftest`).

PS-Q1 VQC - ideal vs official-noise validation

The *same* trained weights are evaluated twice on the 1,500 validation rows: exact statevector (ideal) and the official QFF2026-v2 noise model (noisy). Nothing is retrained or tuned on noisy results. Validation-only; not held-out test evidence.

Official noise configuration

- Helper `noise_model/benchmark_noise_model.py`, imported unchanged; version **QFF2026-v2**: single-qubit depolarizing 0.01, two-qubit depolarizing 0.05, symmetric readout error 0.02; simulator `noisy_simulator(seed)`.
- **Transpilation: `prepare_for_noisy_run()` is applied to every executed circuit** (each validation sample's bound circuit; basis rx, ry, rz, cx; optimisation level 1; `seed_transpiler=42`); the helper itself asserts that every gate carries noise.
- Shots: 8192 per circuit (helper default and Task A's value; no PS-Q1 value is documented). Simulator seeds [42, 43, 44, 45, 46]; the seed-42 run is the primary number and the five-seed mean +- sd shows shot-noise variability. `<Z0> = (n0 − n1)/shots` from one classical bit mapped from qubit 0.

Verification evidence (canonical model)

check	result
Noise model attached to simulator	True (noise instructions: cx, cz, h, measure, rx, ry, rz, s, sdg, swap, sx, t, tdg, x, y, z)
Official parameters	p1q=0.01, p2q=0.05, readout=0.02, version QFF2026-v2
Bit order: X on q0 ->	-1.000 (expected -1): ok
Bit order: X on q1 ->	+1.000 (expected +1): ok
Readout noise on an empty circuit, P(1)	0.0181 vs 0.02 (SE 0.0015): ok
Qiskit logical circuit vs PennyLane (exact statevector, 300 val samples)	max |diff| 1.2e-15: ok
Officially transpiled circuit vs PennyLane (exact statevector)	max |diff| 1.1e-15: ok
Noiseless shots vs exact	max |diff| 0.0284 (shot sigma 0.0110): ok
Noisy vs exact on the same subset	mean |z| 0.5389 -> 0.2134

Overall: **all checks passed**.

Ideal vs noisy macro-F1

model	CX	ideal	noisy (seed 42)	noisy mean +- sd (5 sim seeds)	change	noisy recall 0	noisy recall 1	ROC-AUC ideal	ROC-AUC noisy	attenuation slope	corr(ideal, noisy)	decision threshold in	flipped predictions
canonical: scaled, 2 layer(s)	16	0.9993	0.4731	0.4663 +- 0.0048	-0.5262	0.1924	1.0000	1.0000	1.0000	0.396	0.9977	-0.375	0.483
ablation: direct, 2 layer(s)	16	0.9633	0.7993	0.7971 +- 0.0037	-0.1640	0.6652	1.0000	0.9921	0.9911	0.380	0.9968	-0.219	0.196
diagnostic: scaled, 1 layer	8	0.9667	0.9487	0.9496 +- 0.0013	-0.0180	0.9255	0.9867	0.9940	0.9938	0.599	0.9996	-0.196	0.035

Canonical model, ideal confusion matrix:

true \ pred	0 Normal	1 Arrhythmia
0 Normal	898	1
1 Arrhythmia	0	601

Canonical model, noisy confusion matrix (seed 42):

true \ pred	0 Normal	1 Arrhythmia
0 Normal	173	726
1 Arrhythmia	0	601

Canonical per class, ideal:

class	precision	recall	F1
0 Normal	1.0000	0.9989	0.9994
1 Arrhythmia	0.9983	1.0000	0.9992

Canonical per class, noisy (seed 42):

class	precision	recall	F1
0 Normal	1.0000	0.1924	0.3228
1 Arrhythmia	0.4529	1.0000	0.6234

Predicted P(class 1), ideal:

mean	std	min	p5	p25	p50	p75	p95	max	frac predicted 1
0.3910	0.4406	0.0111	0.0136	0.0197	0.0430	0.9278	0.9681	0.9998	0.4013

Predicted P(class 1), noisy (seed 42):

mean	std	min	p5	p25	p50	p75	p95	max	frac predicted 1
0.7082	0.1947	0.4369	0.4835	0.5289	0.6116	0.9350	0.9526	0.9938	0.8847

Why noise changes macro-F1

- **Attenuation, not scrambling.** Under QFF2026-v2 the measured `<Z0>` of the canonical model is close to a scaled copy of the ideal value (slope 0.396, correlation 0.9977), and the ranking of samples is largely preserved (ROC-AUC 1.0000 ideal vs 1.0000 noisy). Depolarizing errors on 16 CX gates (5% each), single-qubit gates (1%) and 2% readout flips shrink the expectation value toward zero.
- **Fixed decision rule.** The trained readout puts the decision boundary at `<Z0> = −0.375`. That threshold was learned for the unattenuated signal; after shrinkage many samples of one class cross it, so the damage is class-specific (noisy recall: class 0 0.192, class 1 1.000). The information largely survives; the calibration of the decision rule does not.
- **Depth trade-off.** The 1-layer scaled model (8 CX) keeps more signal under noise (attenuation slope 0.599, noisy macro-F1 0.9496) but has a lower ideal score (0.9667) and, as shown in the training report, cannot use PC2. The canonical 2-layer model is best ideally and most fragile under noise.
- The direct-angle ablation degrades from 0.9633 to 0.7971.

Circuit resources under the official transpilation

model	depth	1q gates	CX	ops outside basis	un-noised gates	circuits	shots (5 seeds)	transpile time (s)	run time per seed (s)
canonical: scaled, 2 layer(s)	21.0 (21-21)	16.0	16-16	none	none	1500	61,440,000	2.6	8.8
ablation: direct, 2 layer(s)	21.0 (21-21)	16.0	16-16	none	none	1500	61,440,000	2.6	8.8
diagnostic: scaled, 1 layer	11.0 (11-11)	8.0	8-8	none	none	1500	61,440,000	2.4	8.1

Logical circuits: 1 layer depth 11 / CX 8; 2 layers depth 21 / CX 16. Transpilation merges the encoding `RY` with the first trainable `RY` on each qubit (fewer single-qubit gates than the logical circuit) and adds no CX. No coupling map is applied (basis only), so no SWAPs are included here; mapping onto Processor D is part of Task B.

Correction made during the audit

Earlier versions transpiled one *parametrised* template and bound parameters afterwards. With symbolic parameters the transpiler cannot merge the encoding `RY` with the first trainable `RY`, so 4 extra noisy gates per circuit were executed and noisy macro-F1 was understated (direct 2-layer model: 0.778 with the template method vs 0.797 per-circuit, 5-seed means). The pipeline now runs the official `prepare_for_noisy_run()` on every executed circuit, as the rules state.

Blockers

None.

Caveats

- Noisy numbers include shot noise (SE of each `<Z0>` about 0.0110); variability over 5 simulator seeds is reported.
- Ideal numbers are exact, so ideal-vs-noisy differences mix device noise and finite-shot effects.
- The decision threshold is the one learned in ideal training; no noise-aware recalibration is applied in this workstream.

Classical baseline results (teammate)

Classical Logistic Regression Baseline Results: PS-Q1 ECG Biosignal Classification

Problem Statement: PS-Q1 — ECG Biosignal Classification (QML-BIO-MITBIH)
Project Root: `C:\Users\vchar\Documents\Codex\2026-10-29\re\work\QFF2026_QML_Track\QFF2026_QML_Challenge_Kit`
Pipeline: `StandardScaler(fit on train only) → LogisticRegression(solver='lbfgs', max_iter=5000, random_state=42)`
Evaluation Target: `data/q1/ecg_val.csv` (1,500 unseen validation heartbeats)
Official Metric: Macro-averaged F1 Score

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1. Executive Summary

As required by the QFF2026 PS-Q1 specifications, a leakage-safe classical control model was constructed and evaluated prior to any quantum modelling.

Verified Dataset Inventory & Class Balance:

The training partition is strictly loaded from the verified official file:

- **File Path**: `data/q1/ecg_train.csv`
- **Total Rows**: 6,000 labelled samples
- **Class 0 (Normal) Count**: **3,616** (60.27%)
- **Class 1 (Arrhythmia) Count**: **2,384** (39.73%)
- **SHA-256 Checksum**: `c917975279e138c927115f832988f468542a329dcf1399c6b88abbfd13fe9573`
- **Validation File (`data/q1/ecg_val.csv`)**: 1,500 samples (899 Class 0, 601 Class 1; SHA-256: `af9c2c750f80c22f087d8844e924b3e30449b9671aafb0187aa896f6e7b5b85a`)

> [!IMPORTANT] The verified ground-truth training counts are strictly **3,616 Class-0** and **2,384 Class-1** samples. Any prior erroneous reference claiming 3,598 Class-0 and 2,402 Class-1 samples is superseded and corrected.

Ten systematic hyperparameter configurations spanning inverse regularisation strengths $C \in [0.01, 0.1, 1.0, 10.0, 100.0]$ and class weighting options `class_weight` $\in [\text{None}, \text{'balanced'}]$ were trained strictly on `data/q1/ecg_train.csv` (6,000 samples) and validated on `data/q1/ecg_val.csv` (1,500 samples).

Key Baseline Findings:

1. **Perfect Classification Performance Across All Configurations**: Every tested configuration achieved **1.000000 Macro-F1**, **1.000000 Accuracy**, **0 False Positives**, and **0 False Negatives** on the 1,500 validation samples ($\text{TN}=899, \text{FP}=0, \text{FN}=0, \text{TP}=601$).
2. **Rapid Numerical Convergence**: The L-BFGS solver converged in **10 to 14 iterations** across all configurations, well below the 5,000 iteration ceiling.
3. **Selected Model Configuration**: **$C=1.0$, `class_weight=None`** (with $C=0.01$ and $C=0.1$ as strongly regularised equivalent controls).
4. **Root Cause of Perfect Performance**: The QML-BIO-MITBIH dataset features are linearly separable. In particular, the feature `w5_mean` (mean amplitude in the terminal ECG window, samples 140–174) exhibits a natural, non-overlapping threshold gap between Normal and Arrhythmia classes in both training and validation sets.

---

2. Complete Hyperparameter Evaluation Matrix

All configurations were evaluated under fixed seed `random_state=42` and `max_iter=5000` using scikit-learn's default L-BFGS solver:

C	`class_weight`	Convergence Status	Train Macro-F1	Val Macro-F1	Val Accuracy	Class 0 (Normal) Precision / Recall / F1	Class 1 (Arrhythmia) Precision / Recall / F1	Confusion Matrix (TN, FP, FN, TP)
0.01	`None`	Converged (12 iters)	1.000000	**1.000000**	1.000000	1.0 / 1.0 / 1.0	1.0 / 1.0 / 1.0	`[[899, 0], [0, 601]]`
0.10	`None`	Converged (14 iters)	1.000000	**1.000000**	1.000000	1.0 / 1.0 / 1.0	1.0 / 1.0 / 1.0	`[[899, 0], [0, 601]]`

C	`class_weight`	Convergence Status	Train Macro-F1	Val Macro-F1	Val Accuracy	Class 0 (Normal) Precision / Recall / F1	Class 1 (Arrhythmia) Precision / Recall / F1	Confusion Matrix (TN, FP, FN, TP)
1.00	`None`	Converged (11 iters)	1.000000	**1.000000**	1.000000	1.0 / 1.0 / 1.0	1.0 / 1.0 / 1.0	`[[899, 0], [0, 601]]`
10.00	`None`	Converged (11 iters)	1.000000	**1.000000**	1.000000	1.0 / 1.0 / 1.0	1.0 / 1.0 / 1.0	`[[899, 0], [0, 601]]`
100.00	`None`	Converged (11 iters)	1.000000	**1.000000**	1.000000	1.0 / 1.0 / 1.0	1.0 / 1.0 / 1.0	`[[899, 0], [0, 601]]`
0.01	`balanced`	Converged (12 iters)	1.000000	**1.000000**	1.000000	1.0 / 1.0 / 1.0	1.0 / 1.0 / 1.0	`[[899, 0], [0, 601]]`
0.10	`balanced`	Converged (13 iters)	1.000000	**1.000000**	1.000000	1.0 / 1.0 / 1.0	1.0 / 1.0 / 1.0	`[[899, 0], [0, 601]]`
1.00	`balanced`	Converged (10 iters)	1.000000	**1.000000**	1.000000	1.0 / 1.0 / 1.0	1.0 / 1.0 / 1.0	`[[899, 0], [0, 601]]`
10.00	`balanced`	Converged (11 iters)	1.000000	**1.000000**	1.000000	1.0 / 1.0 / 1.0	1.0 / 1.0 / 1.0	`[[899, 0], [0, 601]]`
100.00	`balanced`	Converged (11 iters)	1.000000	**1.000000**	1.000000	1.0 / 1.0 / 1.0	1.0 / 1.0 / 1.0	`[[899, 0], [0, 601]]`

Selected Configuration:

- **Regularisation Strength (C)**: `1.0` (standard L2 penalty)
- **Class Weighting (`class_weight`)**: `None`
- **Solver**: `lbfgs`
- **Max Iterations**: `5000`
- **Random Seed**: `42`
- **Validation Macro-F1**: `1.000000`
- **Validation Accuracy**: `1.000000`

> [!NOTE] Because validation macro-F1 was identical across all grid points, Occam's Razor was applied: the standard default regularisation parameter $C=1.0$ with unweighted loss was selected. For applications desiring maximum regularisation margins against out-of-distribution drift, $C=0.01$ (with weight norm $|\mathbf{w}|_2 = 1.5250$) provides an equally valid candidate.

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3. Engineering & Clinical Analysis

3.1 What Macro-F1 Measures

In binary classification, standard accuracy treats all correct classifications equally: $$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$ When class prevalence is uneven (in `ecg_train.csv`: 3,616 Normal [60.27%] vs. 2,384 Arrhythmia [39.73%]; in `ecg_val.csv`: 899 Normal [59.93%] vs. 601 Arrhythmia [40.07%]), a naive majority-class model predicting only Normal heartbeats would automatically achieve 60.27% training accuracy despite having zero clinical utility.

Macro-averaged F1 computes the unweighted harmonic mean of precision and recall independently for each class, then takes their arithmetic mean: $$F_{1, \text{Normal}} = \frac{2 \cdot \text{Precision}_0 \cdot \text{Recall}_0}{\text{Precision}_0 + \text{Recall}*0}, \quad F*{1, \text{Arrhythmia}} = \frac{2 \cdot \text{Precision}_1 \cdot \text{Recall}_1}{\text{Precision}*1 + \text{Recall}*1}$$ $$\text{Macro-F1} = \frac{F*{1, \text{Normal}} + F*{1, \text{Arrhythmia}}}{2}$$ Macro-F1 forces the model to perform equally well on both minority and majority classes.

3.2 Clinical Importance of Both Normal and Arrhythmia Performance

In automated electrocardiogram (ECG) rhythm triage:

- **False Negatives (FN)**: An arrhythmia heartbeat is misclassified as normal. This carries high clinical risk, as life-threatening ventricular arrhythmias, premature ventricular contractions, or conduction blocks would be missed.
- **False Positives (FP)**: A normal heartbeat is flagged as arrhythmic. This triggers false alarms in telemetry wards, clinical alarm fatigue, and unnecessary invasive follow-up diagnostics.

In our evaluation on the 1,500 validation beats, the classical baseline produced:

- $\text{FP} = 0$ (Precision for Arrhythmia = 1.0000)
- $\text{FN} = 0$ (Recall for Arrhythmia = 1.0000)

3.3 Did Class Weighting Help?

Because the dataset is mildly imbalanced ($1.517 : 1$ ratio in training: 3,616 Class 0 vs. 2,384 Class 1), `class_weight='balanced'` was tested to apply inverse frequency weights ($w_j = \frac{N}{2 \cdot N_j}$).
Because the underlying feature space has an unobstructed geometric separation gap, the unweighted model already separated both classes with wide margins. As a result, `class_weight='balanced'` achieved the exact same 1.000000 Macro-F1 and converged in nearly identical iteration counts ($10–13$ iters vs. $11–14$ iters).

3.4 Convergence Analysis

All 10 configurations converged rapidly in under 15 iterations. The optimization landscape is strictly convex with no saddle points or ill-conditioned curvature under standardised inputs.

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4. Verification Against "Suspiciously Perfect" Results

A score of $100%$ on a clinical biosignal benchmark naturally demands rigorous scrutiny to rule out data leakage, circular evaluation, or target contamination. We conducted five dedicated forensic checks:

Check 1: Preprocessing & Scaler Scope (Strict No-Leakage Protocol)

- Verified that `StandardScaler` was fit strictly on `X_train` (`X_train_scaled = scaler.fit_transform(X_train)`).
- Validation data was strictly transformed out-of-sample (`X_val_scaled = scaler.transform(X_val)`).
- Zero validation information leaked into the scaling parameters (μ, σ).
- **Official Leakage Compliance**: “The test feature file was not used for scaler fitting, model training, hyperparameter tuning, threshold selection, or metric calculation. Any schema or sample_id inspection must be clearly distinguished from modelling use.”

Check 2: Sample Overlap Check

- Checked for shared feature vectors between `ecg_train.csv` and `ecg_val.csv`.
- Result: **0 shared samples**. The partitions are completely disjoint.

Check 3: Decision Margin Analysis

We extracted the continuous signed decision function values $y_i \cdot (\mathbf{w}^T \mathbf{x}_i + b)$ on the validation set:

- **Minimum signed margin on validation**: $+4.8412$ (every single sample lies far on the correct side of the decision boundary).
- **Median signed margin**: $+10.2555$.
- **95th percentile margin**: $+16.8509$. The decision boundary does not graze any validation sample; it sits in a broad, empty geometric corridor.

Check 4: Feature-Level Linearity Audit

To understand why the margin is so large, we analyzed the raw distributions of individual features. As documented in `reports/data_audit.md`, feature **`w5_mean`** (the mean voltage amplitude in the 5th segment of the heartbeat, samples 140–174) has a clean, natural separation gap between Normal and Arrhythmia classes:

- **Train Class 0 range**: $[0.09385, 0.17219]$ mV
- **Train Class 1 range**: $[-0.05071, 0.04422]$ mV
- **Validation Class 0 range**: $[0.10010, 0.16920]$ mV
- **Validation Class 1 range**: $[-0.03777, 0.03271]$ mV

There is an absolute gap of ≈ 0.050 mV between the highest Arrhythmia sample and the lowest Normal sample. A single threshold on `w5_mean` alone achieves near-perfect classification; combining all 30 features provides redundant linear separability.

Check 5: Alignment with Organizer Documentation

In the organizer's locked problem statement document (`BIOSIGNAL_Q1_LOCKED_RULES.md`, line 57), it is explicitly noted:

> "*Include honest limitations: Q1 may be linearly separable, so quantum superiority must not be claimed without evidence.*"

This confirms that the linear separability observed is an intentional property of the QFF2026 PS-Q1 benchmark dataset.

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5. Artifact Index for this Stage

1. **Audit Documentation**: `reports/data_audit.md`
2. **Training Script**: `scripts/train_classical_baseline.py`
3. **Full Metrics CSV**: `reports/classical_baseline_metrics.csv`
4. **Baseline Report**: `reports/classical_baseline_results.md`

Classical data audit (teammate)

QFF2026 PS-Q1 Data Audit: ECG Biosignal Classification

Problem Statement: PS-Q1 — ECG Biosignal Classification (QML-BIO-MITBIH)
Project Root: `C:\Users\vchar\Documents\Codex\2026-10-29\re\work\QFF2026_QML_Track\QFF2026_QML_Challenge_Kit`
Audit Date: October 30, 2026
Auditor: Senior Machine Learning Engineer
Objective: Comprehensive data integrity, schema consistency, and leakage-safe verification across official ECG partitions.

---

1. File Inventory and Paths

All dataset partitions reside in the designated official path under `data/q1/` relative to the challenge kit root:

Dataset Partition	File Path	File Size (Bytes)	SHA-256 Checksum
Training Partition	`data/q1/ecg_train.csv`	3,606,415	`c917975279e138c927115f832988f468542a329dcf1399c6b88abbfd13fe9573`
Validation Partition	`data/q1/ecg_val.csv`	901,871	`af9c2c750f80c22f087d8844e924b3e30449b9671aafb0187aa896f6e7b5b85a`
Test Features Partition	`data/q1/ecg_test_features.csv`	870,532	`aadbe112b6a158ef74ddbdd1902cb7fd3cb8e565f42a635d77d7ddc3bf4fe312`

> [!NOTE] All files match the official challenge distribution bit-for-bit with verified SHA-256 cryptographic hashes. The source files in `Downloads` were verified as identical copies and left unmodified.

---

2. Dataset Dimensions and Schema Verification

2.1 Row and Column Counts

Partition	Total Rows	Total Columns	Feature Columns	Target / Identifier Columns
`ecg_train.csv`	6,000	31	30 (`w1_mean` … `w5_kurt`)	`label` (int64)
`ecg_val.csv`	1,500	31	30 (`w1_mean` … `w5_kurt`)	`label` (int64)
`ecg_test_features.csv`	1,500	31	30 (`w1_mean` … `w5_kurt`)	`sample_id` (string: `s0` … `s1499`)

2.2 Exact Column Names and Ordering

The 30 numeric feature columns represent statistical moments across five 35-sample time-domain windows of 187-sample ECG heartbeats centred on the R-peak:

1. Window 1 (samples 0–34): `w1_mean`, `w1_std`, `w1_min`, `w1_max`, `w1_skew`, `w1_kurt`
2. Window 2 (samples 35–69): `w2_mean`, `w2_std`, `w2_min`, `w2_max`, `w2_skew`, `w2_kurt`
3. Window 3 (samples 70–104): `w3_mean`, `w3_std`, `w3_min`, `w3_max`, `w3_skew`, `w3_kurt`
4. Window 4 (samples 105–139): `w4_mean`, `w4_std`, `w4_min`, `w4_max`, `w4_skew`, `w4_kurt`
5. Window 5 (samples 140–174): `w5_mean`, `w5_std`, `w5_min`, `w5_max`, `w5_skew`, `w5_kurt`

2.3 Train vs. Validation Schema Comparison

- Feature columns match 100% in exact naming, ordering, and data representation between `ecg_train.csv` and `ecg_val.csv`.
- Both training and validation contain exactly `label` as their final column.
- Difference between train and val column lists: **0 columns** (`set(train.columns) ^ set(val.columns) == set()`).

2.4 Test Partition Schema and `sample_id` Verification

- `ecg_test_features.csv` contains `sample_id` as column 0, followed by the exact same 30 feature columns in identical order as train and val.
- `ecg_test_features.csv` contains **no label column**, maintaining strict blind evaluation integrity.
- **Uniqueness**: `sample_id` has exactly 1,500 distinct values (0 duplicates).
- **Ordering**: Values run sequentially from `s0`, `s1`, `s2`, …, up to `s1499`.

- **Exclusion from Modeling**: `sample_id` is an administrative identifier and is explicitly stripped prior to feature scaling and modeling.
- **Strict Leakage Prevention**: “The test feature file was not used for scaler fitting, model training, hyperparameter tuning, threshold selection, or metric calculation. Any schema or sample_id inspection must be clearly distinguished from modelling use.”

---

3. Data Integrity and Quality Checks

Integrity Dimension	Training Partition (`ecg_train.csv`)	Validation Partition (`ecg_val.csv`)	Test Partition (`ecg_test_features.csv`)	Audit Status
Missing / Null Values	0 (0.0%)	0 (0.0%)	0 (0.0%)	**Passed**
Infinite Values (`inf`, `-inf`)	0 (0.0%)	0 (0.0%)	0 (0.0%)	**Passed**
Duplicate Full Rows	0 (0.0%)	0 (0.0%)	0 (0.0%)	**Passed**
Duplicate Feature Vectors	0 (0.0%)	0 (0.0%)	0 (0.0%)	**Passed**
Conflicting Labels on Duplicates	0	0	N/A	**Passed**
Feature Overlap (Train ∩ Val)	0 shared feature vectors	0 shared feature vectors	N/A	**Passed**
Data Types	All 30 features: `float64` `label`: `int64`	All 30 features: `float64` `label`: `int64`	All 30 features: `float64` `sample_id`: `object` (str)	**Passed**

---

4. Class Balance and Label Distributions

Target label definitions:

- `0`: Normal heartbeat
- `1`: Arrhythmia

Partition	Total Samples	Class 0 (Normal)	Class 1 (Arrhythmia)	Class 0 Ratio	Class 1 Ratio	Imbalance Ratio (0:1)
`ecg_train.csv`	6,000	3,616	2,384	60.27%	39.73%	1.517 : 1
`ecg_val.csv`	1,500	899	601	59.93%	40.07%	1.496 : 1
Combined Labelled	7,500	4,515	2,985	60.20%	39.80%	1.513 : 1

> [!IMPORTANT] The class distributions in training ($60.27%$ vs. $39.73%$) and validation ($59.93%$ vs. $40.07%$) are closely matched ($\Delta < 0.34%$), confirming a clean, stratified partition. The slight class imbalance justifies testing both standard and `class_weight='balanced'` settings in classical modeling.

---

5. Feature Ranges and Descriptive Statistics

Descriptive statistics across the 30 features demonstrate consistent distributional alignment across training and validation partitions:

Feature	Train Mean	Train Std	Train Min	Train Median	Train Max	Val Mean	Val Std	Val Min	Val Median	Val Max
`w1_mean`	0.0273	0.0198	-0.0402	0.0335	0.0712	0.0271	0.0196	-0.0313	0.0334	0.0669
`w1_std`	0.0631	0.0131	0.0299	0.0605	0.1069	0.0631	0.0134	0.0323	0.0602	0.1063
`w1_min`	-0.0807	0.0428	-0.3022	-0.0665	-0.0161	-0.0806	0.0429	-0.2714	-0.0665	-0.0227
`w1_max`	0.1738	0.0506	0.0484	0.1699	0.3738	0.1727	0.0510	0.0507	0.1696	0.3727
`w1_skew`	0.5219	0.4480	-1.4690	0.6230	1.9177	0.5132	0.4501	-1.3889	0.6134	1.4959
`w1_kurt`	-0.2010	0.5487	-1.3634	-0.2866	6.1472	-0.2224	0.5470	-1.4486	-0.3251	3.9538
`w2_mean`	0.0069	0.0145	-0.0423	0.0032	0.0596	0.0068	0.0145	-0.0353	0.0032	0.0599
`w2_std`	0.0502	0.0321	0.0176	0.0334	0.1485	0.0501	0.0322	0.0192	0.0332	0.1524
`w2_min`	-0.0843	0.0406	-0.3661	-0.0710	-0.0204	-0.0841	0.0412	-0.2902	-0.0709	-0.0262
`w2_max`	0.1445	0.1590	0.0269	0.0800	0.7036	0.1435	0.1597	0.0265	0.0795	0.7052
`w2_skew`	0.3318	0.9411	-1.8223	0.0879	3.4741	0.3208	0.9439	-1.0915	0.0788	3.4011
`w2_kurt`	0.7536	2.7576	-1.4176	-0.1957	12.7041	0.7345	2.7879	-1.3243	-0.2221	12.1766
`w3_mean`	0.1152	0.2725	-0.0356	-0.0092	0.9392	0.1099	0.2630	-0.0283	-0.0094	0.9330

Feature	Train Mean	Train Std	Train Min	Train Median	Train Max	Val Mean	Val Std	Val Min	Val Median	Val Max
w3_std	0.1765	0.1103	0.0562	0.1220	0.4823	0.1760	0.1080	0.0609	0.1223	0.4671
w3_min	-0.1839	0.1300	-0.3877	-0.2320	0.2608	-0.1850	0.1287	-0.3964	-0.2320	0.2318
w3_max	0.6130	0.3708	0.0536	0.4644	1.6763	0.6100	0.3609	0.0976	0.4673	1.6715
w3_skew	0.9990	0.5976	-1.1409	1.0913	2.5746	1.0220	0.5828	-1.2058	1.1143	2.4674
w3_kurt	3.5201	2.7923	-1.4529	4.5895	9.9698	3.5382	2.7908	-1.4450	4.5973	9.8392
w4_mean	0.1103	0.0579	-0.0380	0.1284	0.3001	0.1091	0.0568	-0.0173	0.1269	0.2981
w4_std	0.2497	0.1177	0.0270	0.2951	0.4226	0.2477	0.1166	0.0277	0.2913	0.4086
w4_min	-0.1509	0.0472	-0.3026	-0.1568	-0.0378	-0.1498	0.0474	-0.2873	-0.1557	-0.0248
w4_max	0.7547	0.3547	0.0358	0.8869	1.3379	0.7506	0.3517	0.0372	0.8766	1.2498
w4_skew	1.2885	0.5600	-1.4783	1.5682	2.6476	1.2962	0.5619	-0.9530	1.5676	2.6514
w4_kurt	0.8573	0.7340	-1.2915	0.9790	7.0362	0.8863	0.8201	-1.1822	0.9858	6.8962
w5_mean	0.0804	0.0670	-0.0507	0.1161	0.1722	0.0799	0.0665	-0.0378	0.1153	0.1692
w5_std	0.0880	0.0296	0.0266	0.0960	0.1424	0.0879	0.0292	0.0269	0.0958	0.1402
w5_min	-0.0649	0.0544	-0.3408	-0.0451	0.0219	-0.0662	0.0554	-0.3641	-0.0461	0.0203
w5_max	0.2580	0.1189	0.0364	0.3040	0.4740	0.2572	0.1183	0.0416	0.3046	0.4662
w5_skew	0.2477	0.3166	-1.3710	0.3405	1.6198	0.2328	0.3353	-1.8863	0.3310	1.3525
w5_kurt	-0.6943	0.6363	-1.5388	-0.9270	5.2792	-0.6893	0.6235	-1.4339	-0.9128	4.5646

Critical Finding on Linear Separability:

A detailed class-by-class audit revealed that the feature **`w5_mean`** (the arithmetic mean amplitude of the 5th segment, samples 140–174) exhibits a strict, non-overlapping threshold gap between Normal heartbeats (`label = 0`) and Arrhythmia heartbeats (`label = 1`):

- **Training Set (`ecg_train.csv`)**:
  - Class 0 (`w5_mean`): range $[0.09385, 0.17219]$ (mean: 0.1335, std: 0.0161)
  - Class 1 (`w5_mean`): range $[-0.05071, 0.04422]$ (mean: -0.0003, std: 0.0102)
  - Linear Separation Gap: **0.04963 mV**
- **Validation Set (`ecg_val.csv`)**:
  - Class 0 (`w5_mean`): range $[0.10010, 0.16920]$ (mean: 0.1331, std: 0.0161)
  - Class 1 (`w5_mean`): range $[-0.03777, 0.03271]$ (mean: 0.0003, std: 0.0097)
  - Linear Separation Gap: **0.06739 mV**

This confirms that the problem is natively linearly separable along the 5th window amplitude dimension, directly explaining the 1.0000 accuracy of the linear classifier.

---

6. Formal Audit Checklist

- [x] **File Paths**: All files are located in `data/q1/` at the official project root.
- [x] **Row Counts**: Train = 6,000, Val = 1,500, Test = 1,500.
- [x] **Column Counts**: Train = 31, Val = 31, Test = 31.
- [x] **Exact Feature Names**: Exactly the 30 features `w1_mean` through `w5_kurt` in identical order.
- [x] **Data Types**: All features are numeric (`float64`), labels are integer (`int64`), test `sample_id` is string.
- [x] **Missing / Infinite Values**: Zero NaN, zero null, zero inf values across all partitions.
- [x] **Duplicate Checks**: Zero duplicate rows, zero duplicate feature vectors, zero label conflicts.
- [x] **Train/Val Schema Match**: Exact 100% column match.
- [x] **Test Identifier Verification**: `sample_id` contains `s0` through `s1499` in exact sequential order.
- [x] **Leakage Prevention**: Scalers must be fit strictly on `ecg_train.csv`.
- [x] **Data Quality Verdict**: **CLEAN AND APPROVED FOR MODELING**.

Classical model specification (teammate)

Frozen Classical Baseline Specification: PS-Q1 ECG Biosignal Classification

Problem Statement: PS-Q1 — ECG Biosignal Classification (`QML-BIO-MITBIH`)
Project Root: `C:\Users\vchar\Documents\Codex\2026-10-29\re\work\QFF2026_QML_Track\QFF2026_QML_Challenge_Kit`
Status: **FROZEN & LOCKED**
Date Locked: October 30, 2026
Artifact Path: `artifacts/classical_q1_pipeline.joblib`
Inference Script: `scripts/predict_classical_q1.py`

---

1. Model Architecture and Hyperparameter Configuration

The selected classical control model is a scikit-learn `Pipeline` consisting of an out-of-sample `StandardScaler` followed by regularised `LogisticRegression`:

```
Pipeline(steps=[
    ('scaler', StandardScaler(copy=True, with_mean=True, with_std=True)),
    ('classifier', LogisticRegression(
        C=1.0,
        class_weight=None,
        dual=False,
        fit_intercept=True,
        intercept_scaling=1,
        l1_ratio=None,
        max_iter=5000,
        multi_class='auto',
        n_jobs=None,
        penalty='l2',
        random_state=42,
        solver='lbfgs',
        tol=0.0001,
        verbose=0,
        warm_start=False
    ))
])
```

Parameter Summary:

- **Pipeline Structure**: `StandardScaler` $\to$ `LogisticRegression`
- **Inverse Regularisation Strength (C)**: `1.0` (standard L2 ridge penalty)
- **Class Weighting (`class_weight`)**: `None` (unweighted cross-entropy loss)
- **Optimization Solver**: `lbfgs` (Limited-memory Broyden–Fletcher–Goldfarb–Shanno)
- **Maximum Solver Iterations (`max_iter`)**: `5000`
- **Tolerance (`tol`)**: `1e-4`
- **Random State (`random_state`)**: `42`
- **Convergence Information**: Converged successfully in **11 iterations** (well below the 5,000 threshold).

---

2. Dataset Scope and Scaler Scope

File Paths and Checksums:

- **Training Data**: `data/q1/ecg_train.csv`
  - Total Samples: 6,000
  - Target Distribution: Class 0 (Normal) = **3,616** ($60.27%$), Class 1 (Arrhythmia) = **2,384** ($39.73%$)
  - SHA-256 Checksum: `c917975279e138c927115f832988f468542a329dcf1399c6b88abbfd13fe9573`
- **Validation Data**: `data/q1/ecg_val.csv`
  - Total Samples: 1,500

- Target Distribution: Class 0 (Normal) = **899** ($59.93%$), Class 1 (Arrhythmia) = **601** ($40.07%$)
- SHA-256 Checksum: `af9c2c750f80c22f087d8844e924b3e30449b9671aafb0187aa896f6e7b5b85a`

Scaler Scope:

The `StandardScaler` was fitted **strictly and exclusively on `data/q1/ecg_train.csv`** (6,000 samples). Scaling parameters ($\mu \in \mathbb{R}^{30}, \sigma \in \mathbb{R}^{30}$) were computed solely on the training partition: $$\mathbf{z} = \frac{\mathbf{x} - \boldsymbol{\mu}_{\text{train}}}{\boldsymbol{\sigma}_{\text{train}}}$$ Validation samples were transformed out-of-sample using these frozen training statistics.

Explicit Confirmation of No Test Leakage:

> "The test feature file was not used for scaler fitting, model training, hyperparameter tuning, threshold selection, or metric calculation. Any schema or sample_id inspection must be clearly distinguished from modelling use."

`data/q1/ecg_test_features.csv` was never loaded, inspected, or processed during the fitting of this pipeline.

---

3. Verification & Evaluation Metrics

The frozen model was independently evaluated across training and unseen validation splits:

Evaluation Metric	Training Split (`ecg_train.csv`, N=6,000)	Validation Split (`ecg_val.csv`, N=1,500)	Target Criterion
Macro-Averaged F1	**1.000000**	**1.000000**	Official Challenge Metric
Accuracy	**1.000000**	**1.000000**	$100.0%$
True Negatives (TN)	3,616	899	Exact Match
False Positives (FP)	0	0	0 False Alarms
False Negatives (FN)	0	0	0 Missed Arrhythmias
True Positives (TP)	2,384	601	Exact Match

Class-Wise Validation Breakdown:

- **Class 0 (Normal Heartbeat, Support = 899)**:
  - Precision: `1.000000`
  - Recall: `1.000000`
  - F1-Score: `1.000000`
- **Class 1 (Arrhythmia, Support = 601)**:
  - Precision: `1.000000`
  - Recall: `1.000000`
  - F1-Score: `1.000000`

Confusion Matrix on Validation:

$$\mathbf{C} = \begin{bmatrix} \text{TN} & \text{FP} \\ \text{FN} & \text{TP} \end{bmatrix} = \begin{bmatrix} 899 & 0 \\ 0 & 601 \end{bmatrix}$$

---

4. Inference & Verification Protocol

The standalone inference script `scripts/predict_classical_q1.py` provides a certified interface to the serialized pipeline:

1. **Format Compliance**: Output schema strictly generates `sample_id,label` with binary integer labels ${0, 1}$, fully passing `tools/validate_predictions.py`.
2. **Feature Integrity**: Validates the presence of all 30 official statistical features (`w1_mean` through `w5_kurt`) in canonical order and rejects unexpected schema inputs.
3. **Execution Guard**: By default, running `python scripts/predict_classical_q1.py` executes in dry-run/in-memory verification mode and **does not write test predictions to disk** unless an explicit destination path (`--output <path>`) is provided.

---

5. Decision on Final Test Predictions

> [!IMPORTANT] **Final test predictions (`predictions_q1.csv`) are intentionally postponed.**
> Per competition guidelines and scientific protocol, `predictions_q1.csv` will not be finalized or written to the project

root until the Quantum Variational Classifier (VQC) has been constructed, trained, noise-tested, and evaluated in direct comparison with this classical baseline.

Audit notes (independent review)

Everything below was reproduced by `work/audit_checks.py` (seeded; read-only on `results/` and `data/`). Environment used: the repo's own pins, Python 3.12.3. `python main.py` regenerated every file in `results/` byte-identically to the committed versions.

1. What was verified

Check	Result
`main.py` reproduces committed tables	Identical (git diff empty)
Noiseless shot simulation of the transpiled A and B circuits vs statevector (40k shots)	Max difference 0.012 (about 2.4 standard errors over 40 samples), so measurement and parity logic are consistent
Transpiled circuits are separate and valid	A: 42 CX, 5-qubit chain, qubits 0-3 used. B: 12 CX, qubits 2-5 used. Circuits differ. No CX violates the directed coupling map. All gates are in the processor basis
VQC and logistic regression make the same 3 errors	Test indices 10, 11, 36. Index 11 sits at ideal expectation -0.044 (almost on the boundary)
Train vs test accuracy of the ideal VQC	0.875 train, 0.925 test. The test split is slightly easier than train; do not present 92.5% as the expected accuracy
Output strength	Final training MSE 0.605. Mean absolute expectation 0.30, max 0.64 (a perfect classifier would approach 1). 13 of 40 test samples have margin below 0.2
Margin of the closest test samples	Smallest margin 0.029; 2 samples within 0.05 of the threshold

2. Why identical accuracy for Processors A and B is legitimate here

Accuracy is thresholded. Depolarizing and readout noise mostly shrink expectations toward 0, which keeps the sign (the decision) for samples with a healthy margin. At the repo's noise rates (p1q 0.001, pCX 0.005, readout 0.01) over 20 simulator seeds: A accuracy 0.9225 +/- 0.0075, B 0.9250 +/- 0.0000. A differs from B by less than one sample. Physical degradation is still visible: expectation MSE vs ideal 0.0097 (A) vs 0.0030 (B), Brier 0.1719 vs 0.1607 (ideal 0.1487), mean |expectation| 0.214 (A) vs 0.254 (B) vs 0.300 (ideal).

3. The headline caveat: the repo's noise is much weaker than the organizer's

The Phase 2 rules require noise model QFF2026-v2 (p1q 0.01, p2q 0.05, readout 0.02) with circuits transpiled to the fully-noised basis rx/ry/rz/cx. Re-running the same trained VQC and the same coupling maps under it (8 simulator seeds, 8192 shots, 40 test samples):

Setting	CX	Depth	Accuracy	ROC-AUC	Mean abs expectation (ideal 0.300)	Prediction flips vs ideal
Processor A (linear)	42	73	0.816 +/- 0.053	0.869	0.015	5.1 of 40
Processor B	12	30	0.925 +/- 0.013	0.916	0.094	0.2 of 40
All-to-all (no routing)	12	20	0.916 +/- 0.012	0.917	0.118	0.4 of 40

So the "accuracy does not depend on topology" finding holds only in the weak-noise regime. At the official noise level Processor A's signal is almost entirely washed out (mean |expectation| 0.015) and accuracy drops about 11 points. The A-vs-B accuracy gap here is large relative to the seed spread, but it is 40 samples and one trained parameter set.

4. Safe and unsafe claims for Phase 1 material

Safe: accuracy is identical at 40 samples and low noise; A has 3.5x the CX count and about 3.2x the expectation error of B; the first "new data" result (12.5% to 21%) was an out-of-distribution task mismatch, not generalization. Not safe: "topology does not matter", "VQC matches classical" as a general statement (one task, n=40, same 3 errors), "mesh" for B (it is a sparse heavy-hex-inspired graph), any statement about real hardware.

5. Gap against the Phase 2 kit (v3) and rubric

This repository is the Phase 1 deliverable. Against the Phase 2 gates:

Item	Status in repo
`solution.ipynb` with PS-Q1 on ECG data (30 features, 6000/1500/1500)	Missing. Repo uses make_classification, 200 samples, 4 features

Item	Status in repo
Task A: Bell fidelity from XX/YY/ZZ, 8192 shots	Missing (reference noisy F is about 0.880; reproduced here as 0.8801 +/- 0.0026)
Task B: `processor_d_config.json` passing the official benchmark	Missing
`predictions_q1.csv`, `results.md`, README with `pip freeze`	Missing
Official noise model QFF2026-v2 for all noisy runs	Not used. The repo has its own noise model
Pinned environment	Mismatch: repo pins qiskit 2.6.0, numpy 2.5.4, pandas 3.0.6; kit pins qiskit 2.3.0, numpy 2.4.6, pandas 3.1.0 and adds PennyLane 0.44.1

Use a fresh environment from the kit's `requirements.txt` for Phase 2; leave this repo's pins unchanged so Phase 1 stays reproducible.

6. Task B evidence (official `task_b_qft3`, this repo's processors as starting points)

Both Phase 1 processors list each link one way only, and direction is enforced, which costs fidelity.

Design	Native gates	Depth	2q gates	SWAPs	Noisy QFT-3 fidelity
Repo A as-is (one-way, 5q chain)	rx,ry,rz,cx	31	12	1	0.550
Repo B as-is (one-way, 9q)	rx,ry,rz,cx	23	9	0	0.648
Linear 5, two-way	rx,ry,rz,cx	22	12	1	0.604
4-qubit ring, two-way	rx,ry,rz,cx	21	12	1	0.627
4-qubit ring + chord (0-2), two-way	rx,ry,rz,cx	18	9	0	0.668
4-qubit ring + chord (0-2), two-way	rx,ry,rz,h,cx	16	9	0	0.678

The ring-plus-chord design runs the VQC's 4-qubit CX ring and the QFT-3 triangle with zero SWAPs, which gives a rationale tied to the team's own circuit. Adding `h` to the native gates helped slightly. It matches the 3-qubit all-to-all triangle (0.678), which is the ceiling for QFT-3 at 9 two-qubit gates. These are measurements of the benchmark as extracted from the uploaded kit text; re-run `python benchmark/qff_qml_benchmark.py` on your own copy before submitting.

7. Model-quality notes

Retraining on a 75% subset of the training data (validation split taken from train only, test untouched), over 4 initialisation seeds: test accuracy 0.90 to 0.95 at 80 iterations, 0.90 to 0.95 at 300 iterations; MSE plateaus near 0.55. More COBYLA iterations do not help; the limit is the model (parity observable ZZZZ with MSE loss), not the optimiser budget. The 92.5% is plausible, and one sample is 2.5 points.

Remaining weaknesses: 40 test samples (Wilson 95% interval for 37/40 is about 80% to 97%), one synthetic task, no validation split in the original pipeline, ZZZZ is invariant to bit reversal so it cannot detect bit-order bugs, and the probability `(exp+1)/2` is not calibrated.